PACMAN OOP PROJECT REPORT

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OVERVIEW

PROJECT REQUIREMENTS

Requirement Analysis

- Functional Requirements: "Pac-Man must lose a life when colliding with a red ghost,"
- The game must save the highest score to a text file.
- The user can change the difficulty.
- GUI should be implemented in Swing

Classes

Game ✓

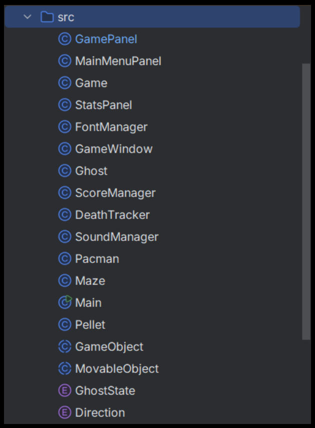
Pacman ✓

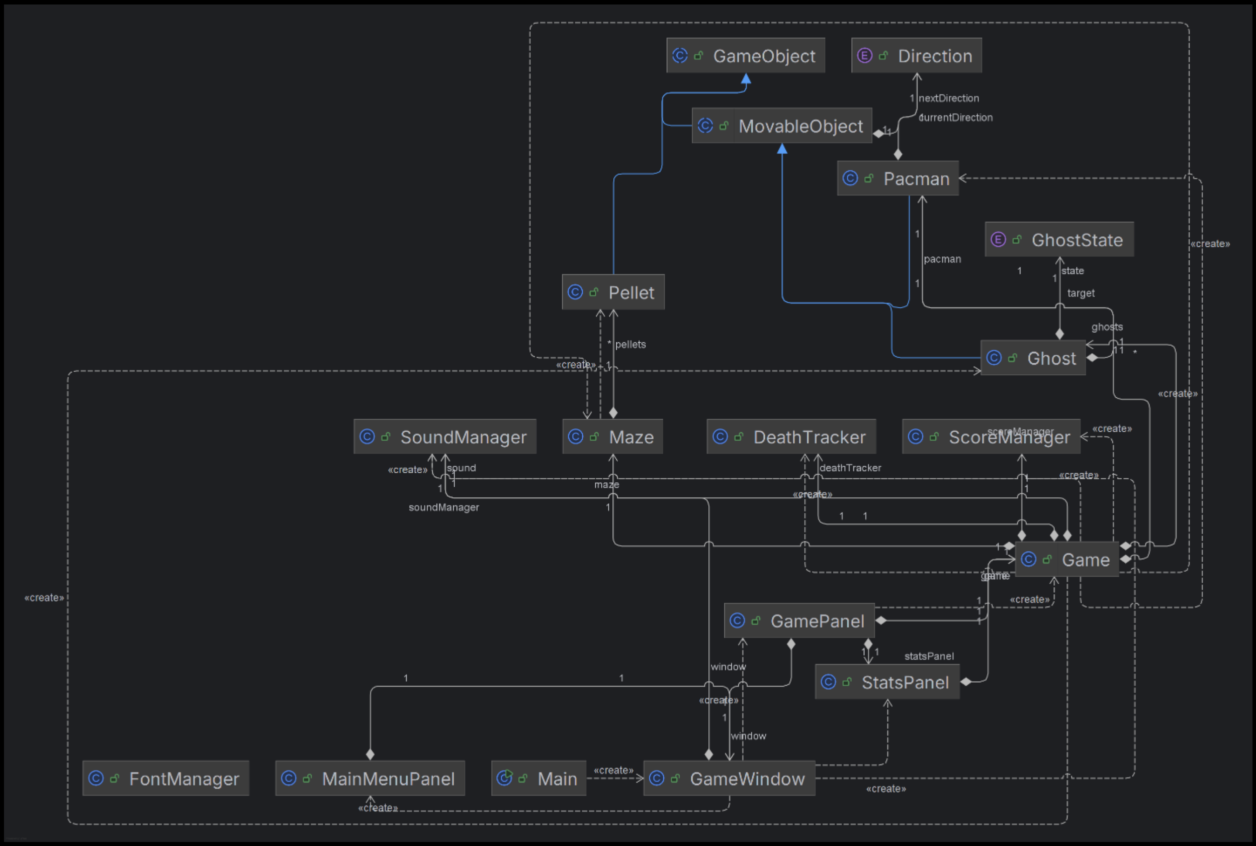
Ghost ✓

Maze ✓

Pellet ✓

ScoreManager ✓





Sequence Diagram

UP arrow key is presswed -> Game detects key press -> Controller updates Pacman direction in the Model -> Model validates the move against the Maze -> Controller tells GamePanel to repaint -> View updates the screen.

Environment

Game Start - Game Exit

We have a main menu for both Starting the Game, Exiting the game, Selecting Game Mode and Selecting Level which will be discussed in later sections

Levels are defined in level folder using Ascii Characters

```
##########################
#...........##...........#
#.####.####.##.####.####.#
#O####.####.##.####.####O#
#........................#
#.####.##.######.##.####.#
#......##...##...##......#
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#O..##......  ......##..O#
###.##.##.######.##.##.###
#B.....##P..##I..##C.....#
##########################
```
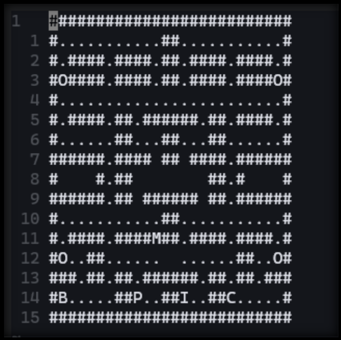

```
#  = wall
M = Pacman Spawn Point
. = Normal Bean
O = Super Pellet (printed as apple fruit)

P = Pinky Ghost
I = Cyan Blue Ghost
C -> Orange Ghost
B -> Red Ghost
```

Walls, Beans, Pacman, Ghost, HighScore

- Walls, Beans, Pacman, Ghosts are all defined in map Ascii representation
- HighScore is saved in `highscore.txt` file



```
##########################
#...........##...........#
#.####.####.##.####.####.#
#O####.####.##.####.####O#
#........................#
#.####.##.######.##.####.#
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#    #.##        ##.#    #
######.## ###### ##.######
#...........##...........#
#.####.####M##.####.####.#
#O..##......  ......##..O#
###.##.##.######.##.##.###
#B.....##P..##I..##C.....#
##########################
```

Maps

as discussed in previous section we can dynamically define maps in levels folder using Ascii Characters

Pacman Style

All assets can be found in Assets folder under main directory of project

Pacman Movement

- you can both control the Pacman using WASD keys or arrow keys
- Pacman can’t bypass the walls
- only moving in allowed directions is allowed

Scoring System

- each key press would lead in one negative score

```java
if (maze.isValidMove(nextX, nextY)) {
    this.x = nextX;
    this.y = nextY;
    this.currentDirection = nextDirection;

    scoreManager.addScore(-1);
}
```

-

- Collecting beans using Pacman would lead in bean deletion

```java
for (Pellet p : maze.getPellets()) {
    if (p.isActive() && p.getX() == pacman.getX() && p.getY() ==
pacman.getY()) {
        eatenPellet = p;
        p.setActive(false);
        scoreManager.addScore(10); // Requirement 4c
```

- Showing Score during the game

- adding 500 scores on collecting all beans

```java
if (maze.getPellets().isEmpty()) {
    System.out.println("YOU WIN! All beans collected.");
    scoreManager.addScore(500); // Requirement 4e
    scoreManager.checkAndSaveHighScore();
    isGameOver = true;
}
```

Ghost

- they can move wisely or randomly based on the difficulty level (on Hard Mode they use A* algorithm and they are nearly unbeatable)
- they can’t bypass walls
- they reroute wisely
- they can collide and kill Pacman

Ending

you should either collide with a normal ghost (not frightened) or collect all beans

You can also pause the game using escape key

Collision

- Pacman and ghosts can’t bypass walls or the border defined in the map
- you would stop moving on collision with walls
- ghosts choose new direction on collision with walls
- beans get deleted on collision with Pacman
- beans would lead in score increasing during the game

Saving Score

we store highest score of the player in highscore.txt file in main project folder

Documentation

How to run the project

you can the project by easily cloning the repository and running mian.java.

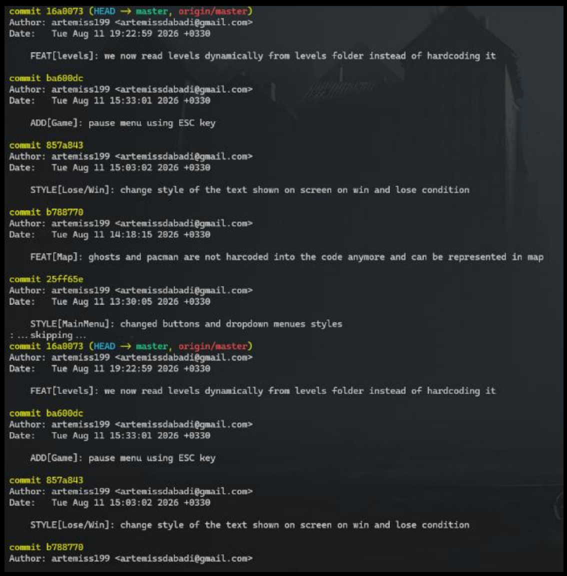
Classes

they are fully described in comments of the source codes

Git

I have used git repeatedly during the project you can see meaningful commits during development

https://github.com/artemiss199/Pacman

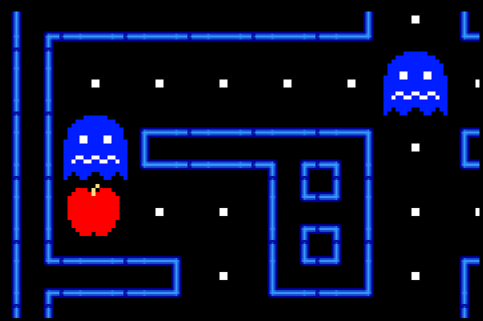


Additional

Power Pellet

power pellets are represented as apples in the game you can eat them and make ghosts frightened

```java
private void checkGhostCollisions() {
    // Pacman hits Ghost (Requirement 6a)
    for (Ghost g : ghosts) {
        if (g.getX() == pacman.getX() && g.getY() ==
pacman.getY()) {
            if (g.getState() == GhostState.FRIGHTENED) {
                soundManager.playSound("eatghost");
                scoreManager.addScore(200);
                g.respawn();
            } else {
```



you can receive 200 scores by eating a frightened ghost

ghosts would jump back to their spawn point immediately after being eaten

Multi Level

you can define you own custom map in levels folder as I did

```
levels
    level0.txt
    level2.txt
    level1.txt
```

```
##########################
#...........##...........#
#.####.####.##.####.####.#
#O####.####.##.####.####O#
#........................#
#.####.##.######.##.####.#
#......##...##...##......#
######.#### ## ####.######
#    #.##        ##.#    #
######.## ###### ##.######
#...........##...........#
#.####.####M##.####.####.#
#O..##......  ......##..O#
###.##.##.######.##.##.###
#B.....##P..##I..##C.....#
##########################
```

```
#  = wall
M = Pacman Spawn Point
. = Normal Bean
O = Super Pellet (printed as apple fruit)

P = Pinky Ghost
I = Cyan Blue Ghost
C -> Orange Ghost
B -> Red Ghost (Blinky)
```

So, you can have as many ghosts as you want

Music

all sound effects can be found in Assets/sfx folder

we have sfx for:

- Main menu
- Game start
- Pacman eating beans
- Pacman eating ghost
- Losing

Advanced Tracking Algorithms

we use A* with a bit of inaccuracy and mis-prediction (dumb acting) based on the difficulty level

if you want to use 100% accuracy, just put the difficulty on Hard in the main menu.

you can find fully explained implementation in Ghost.java file with explanation on the code

```
private void calculateAStarPath(Maze maze) {  1 usage   artemiss199
    int targetX = target.getX();
    int targetY = target.getY();
```

AI Usage

I've mainly used IDE's code completion rather than giving prompt to ai for whole implementation.

Ai helped me on choosing tools for GUI (specially how to render walls better than default dark blue blocks based on their neighbors)

rest of the code like A* implementation or etc. are implemented using help of AI but there is no line of code in the project that was blindly copy pasted and anything that exists in the project is intentional